Earth and Planetary Science Letters, 34 (1977) 125–135
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THE PALAEOMAGNETISM OF SOME UPPER CRETACEOUS KIMBERLITE OCCURRENCES IN SOUTH AFRICA

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Received August 2, 1976
Revised version received November 1, 1976

Palaeomagnetic results are reported from the De Beers, DuToitspan and Wesselton kimberlite pipes (86 ± 3 Ma) at Kimberley and from the Finsch and Koffyfontein pipes. The latter are both less than 150 km from Kimberley and although undated are geologically correlated with the Kimberley pipes. Eighty blocks, oriented without recourse to magnetic methods, have been collected from the five pipes. The samples comprise kimberlite, accidental inclusions of widely varying lithologies and wall rock thus enabling inclusion and baked contact tests to be performed. Extensive alternating field and thermal demagnetization experiments show that the magnetization of the kimberlite is primary. Secular variation is not averaged out in any one pipe (with the possible exception of Finsch) and it is suggested the three Kimberley pipes were emplaced simultaneously. The mean pole position of the five pipes (57.2°E, 58.2°S with $K = 25.8$ and $A_{95} = 15.3°$) is believed to be a good estimate of the palaeomagnetic pole 86 ± 3 Ma ago. It is now possible to state that the African Mesozoic palaeomagnetic pole remained essentially in the same position until at least 86 ± 3 Ma ago.

1. Introduction

There are few reliable palaeomagnetic results from continental Africa from Cretaceous or Early Tertiary rock formations. At the commencement of the present study none were available between those from the Lupata volcanics of Mozambique (110–106 Ma) [1] and the Ethiopian traps [2] which are of Miocene age (20–25 Ma) [3–5]. In 1973 a mean Rb–Sr age of 86 ± 3 Ma was published [6] for the Wesselton, DuToitspan and De Beers kimberlite pipes of Kimberley, South Africa. As kimberlite had previously been shown [7] to be a suitable material for palaeomagnetic work the three pipes were sampled to try to provide an Upper Cretaceous result. It was decided to include the Finsch and Koffyfontein pipes (130 km west and 80 km southeast of Kimberley respectively) in the study because although they were undated the geology indicates that they are post-Karoo and it is likely that their ages are very similar to those of the dated pipes.

2. Geology and sampling

All of the pipes sampled were intruded through the basement complex and the Ventersdorp Supergroup (Proterozoic). Accidental inclusions derived from the upper part of the Karoo System, which is no longer preserved in the area, have been identified in the pipes [8]. Williams concluded from the character of the Karoo inclusions that the full sequence of the Karoo System, including at least some of the Stormberg lava, was present when the pipes were emplaced. Hawthorne [9] has presented a generalized model for a kimberlite pipe prior to erosion and removal of part of the Karoo System and his model displays the geology with sufficient accuracy for all the pipes except Finsch. In the area of the latter, ironstones and dolomites of the Transvaal Supergroup (also Proterozoic) are present in the stratigraphical sequence above the Ventersdorp Supergroup.

To obtain hard unweathered kimberlite (blueground) it is necessary to sample deep underground

TABLE 1

Geographical coordinates, sampling details and cross-section dimensions

Mine	Lat. (°S)	Long. (°E)	Kimberlite	Accidental inclusions	Wall rock	a(m)	b(m)
De Beers (A)	28.7	24.8	11	4	8	240	85
DuToitspan (B)	28.7	24.8	7	10	3	200 *	85
Finsch (C)	28.4	23.5	7	7		420	420
Wesselton (D)	28.7	24.8	2	6	8	300	120
Koffyfontein (E)	29.4	25.0	4	3		105	95

a and b are the length of the major and minor axes of the ellipses to which the pipe cross-sections at the levels sampled have been approximated.

* The true value is not known because the pipe had not been mined throughout the whole cross-section. The figure quoted refers to the mined cross-section and is thought to be a significant underestimate of the true value.

in mines. Oriented block samples must be collected as the use of fuel driven palaeomagnetic coring drills is forbidden. The orientation of blocks by means of a magnetic method (e.g. Brunton compass) is always suspect because of possible magnetic effects of the material actually being sampled. In a mine the proximity of mining equipment makes such a method even more hazardous and it was necessary to develop a non-magnetic method. During mining operations survey pegs are placed in the roof of the tunnels and the direction of the line joining any two of these pegs is determined very accurately by non-magnetic surveying methods. Thus a theodolite was positioned on the line joining two pegs, suitably illuminated, and calibrated by sighting on either of them. A block orienter, suitable for use underground in comparative darkness, was developed [10] and used in conjunction with the theodolite. An accuracy of orientation of the cores drilled from the blocks equal to that obtained by drilling above ground is claimed, i.e. approximately 1°.

Accidental inclusions and wall rock samples were collected where possible as well as kimberlite; the number of each type from each pipe is given in Table 1. The accidental inclusions from the De Beers pipe were all of Ventersdorp lava while those from the other pipes were principally of Karoo origin. Inclusions were collected to help establish the time of origin of the kimberlite magnetization and determine the temperature of emplacement of the kimberlite; the latter study is reported separately [11]. Wall rock samples on, and at various distances from, the contact with the kimberlite were obtained for similar reasons. Those from De Beers and Wesselton were all of Ventersdorp lava while those from DuToitspan were of Archaean granite gneiss.

Samples were collected where possible at different levels and throughout the cross-section of the pipes. Precise positions of samples are given by McFadden [10]. The levels below the surface were: De Beers – 585 m, 595 m, 610 m; Wesselton – 600 m, 615 m; DuToitspan – 755 m; Finsch – 348 m. Koffyfontein is presently worked open-cast (thus the theodolite was calibrated via the sun) and the only suitable kimberlitic material was a hard kimberlite dyke running through the weathered yellow ground. The dyke has been fractured extensively and has expanded since the removal of the yellow ground around it. Samples from this pipe thus might be expected to show an anomalously high scatter of directions of magnetization.

The shape and area of cross-section of the pipes vary from pipe to pipe and with depth within each pipe; in this study the latter variation was small over the levels sampled. If the cross-sections at the levels sampled are approximated to ellipses then the lengths of the major and minor axes give an idea of the size of the pipes. These dimensions are given in Table 1.

3. Measurements and results

3.1. General procedures

Cores were drilled from the block samples and at least two specimens obtained from each sample. Palaeomagnetic measurements were made using

either a PAR-SM1 spinner magnetometer or, where intensities were very low, the BvO spinner magnetometer [12]. At least one specimen from each sample was used for alternating magnetic field (AF) demagnetization experiments using the apparatus described by McElhinny [13]. A further specimen (or specimens) was thermally demagnetized using newly constructed apparatus [10] based on the design of McElhinny et al. [14]. These experiments were performed in an atmosphere of nitrogen and specimens cooled in an external magnetic field of less than 5 nT (5 gammas). AF demagnetization is more likely to introduce small randomizing magnetizations than thermal, hence the results of thermal demagnetization have been taken as definitive and the results of AF demagnetization studies have been used for confirmation.

The results show that the wall rock samples can be grouped broadly into three classes:

Class 1: those whose NRM directions were close to the direction of magnetization of the rock formation from which they were obtained.

Class 2: those whose NRM directions were similar to the kimberlite direction but which had not been totally remagnetized.

Class 3: those which had been totally remagnetized during emplacement of the kimberlite.

The results from classes 1 and 2 are discussed elsewhere [11].

3.2. De Beers Mine

Progressive AF demagnetization of a specimen of kimberlite showed no significant change in its direction of magnetization until subjected to a peak field of 50 mT. At this field the specimen became magnetically unstable. Specimens from the other kimberlite samples were then demagnetized in 5, 10, 15 and 20 mT and in no case was any significant change in direction observed. The mean direction for the kimberlite samples is given in Table 2 for NRM and 20 mT, the latter value being chosen since the estimate of the precision parameter, k [15], was highest. A further pilot specimen was demagnetized at successively higher temperatures to just in excess of the Curie temperature of magnetite when the magnetization became unmeasurable. At no time did the direction of magnetization deviate from the NRM direction. The remaining kimberlite specimens were demagnetized at 400°C and the mean direction is given in Table 2.

Specimens from all four accidental inclusion samples were progressively AF demagnetized and except for those from sample A13 no significant changes in direction occurred. Fig. 1a shows the demagnetization curves and Fig. 2a the directional changes for two specimens from A13. Further specimens from the inclusions were thermally demagnetized and the same type of behaviour was observed; no changes in direction were observed except for the specimens of A13. The stable direction of magnetization revealed in specimens from A13 at the higher temperatures (Fig. 2b) is the same as that revealed by AF demagnetization and is believed to be the primary direction of the inclusion which was not totally remagnetized on emplacement of the kimberlite. Statistics for the remaining three samples are given in Table 2.

Four of the wall rock samples appeared at first to fall into class 3 because no significant changes in direction occurred in any of the specimens on AF demagnetization up to 100 mT. This was also the case on thermal demagnetization except for the two specimens from sample A26 (which was then recognized as class 2). Fig. 3 shows the blocking temperature spectra of the samples consist almost entirely of thermally discrete components [16] with that of A26 being slightly less than the others. Fig. 3 also shows the changes in direction which occurred for specimen A26D (the other behaved similarly) between 500°C and the Curie point of magnetite. Very rapid movement took place along a great circle to a direction very similar to that observed in the Upper Ventersdorp lavas [17,18] of which it is part. Since the magnetization remained stable at the NRM direction until 500°C the directions at 400°C for the two specimens from A26 have been included in the statistics with the other three wall rock samples (Table 2).

The mean direction of the kimberlite samples after AF demagnetization is almost identical to that after thermal demagnetization; the same is true in the case of both accidental inclusions and wall rock samples. The mean directions of the kimberlites and the accidental inclusions cannot be distinguished from each other after either AF or thermal cleaning. Comparison in the usual manner [19,20] of the mean direction of the wall rock samples with that of the kimberlite and accidental inclusions is not possible

TABLE 2

Directions of magnetization of the various sample populations

Sample type	N	NRM				Treatment	After treatment			
		D	I	k	α_{95} (deg.)		D	I	k	α_{95} (deg.)
De Beers Mine										
Kimberlite	11	329.0	−73.0	86.5	4.9	20 mT	331.4	−74.1	92.3	4.8
	11	328.6	−73.4	89.1	4.9	400°C	331.3	−73.3	130.2	4.0
Inclusions	3	337.3	−69.3	182.6	9.1	20 mT	338.0	−70.5	406.2	6.1
	3	338.2	−71.3	212.7	8.5	400°C	334.9	−71.9	133.7	10.7
Class 3 wall rock	4	339.7	−67.8	342.7	5.0	20 mT	338.4	−66.8	380.4	4.7
	4	337.0	−67.2	295.3	5.4	400°C	339.5	−68.7	706.4	3.5
DuToitspan Mine										
Kimberlite	7	337.6	−72.3	98.4	6.1	40 mT	335.2	−72.9	58.2	8.0
	7	339.6	−72.0	115.1	5.7	400°C	334.7	−73.1	93.6	6.3
Inclusions	10	339.6	−67.6	71.6	5.7	40 mT	342.1	−69.0	82.7	5.3
	10	341.5	−66.6	77.6	5.5	400°C	336.5	−66.3	86.7	5.2
Wesselton Mine										
Kimberlite	2	339.2	−64.5	28.2		60 mT	331.1	−68.9	89.3	
	2	342.4	−66.0	31.3		463°C	342.1	−67.4	39.2	
Inclusions	6	337.7	−72.5	113.1	6.3	40 mT	337.4	−73.4	175.0	5.1
	6	336.6	−73.5	127.3	6.0	400°C	334.9	−73.2	131.2	5.9
Finsch Mine										
Kimberlite	7	310.0	−47.6	42.3	9.4	20 mT	312.9	−47.4	116.5	5.6
	7	314.3	−49.2	82.4	6.7	400°C	315.1	−46.2	80.7	6.8
Inclusions	7	6.6	−46.3	1.3		60 mT	15.5	−63.9	1.6	
	7	12.2	−42.6	1.2		500°C	59.4	−14.4	1.0	
Koffyfontein Mine										
Kimberlite	4	359.1	−77.9	75.3	10.7	40 mT	330.1	−85.1	69.4	11.1
	4	352.3	−78.4	47.4	13.5	566°C	12.7	−81.6	68.7	11.2
Inclusions	3	353.6	−65.4	33.3	21.7	40 mT	335.2	−69.4	82.3	13.7
	3	328.5	−71.6	49.7	17.7	400°C	341.9	−67.3	46.2	18.3

because of the very high precision of the wall rock result. All the class 3 wall rock samples were collected within 0.8 m of the kimberlite contact and since the pipe must have cooled radially these samples would have acquired their secondary (kimberlitic) magnetization over a short period of time relative to the period over which different samples scattered within the pipe would have acquired their magnetization. Therefore a physical reason exists for expecting that the wall rock samples only sampled a portion of the secular variation sampled by the pipe itself and hence for expecting that the precision of the wall rock result should be significantly higher than the result obtained from samples scattered within the pipe.

However, there is considerable overlap of the circles of confidence for the kimberlite and wall rock directions and the circle of confidence of the latter is contained entirely within that for the accidental inclusions. (In addition it can be shown statistically [10] that the mean direction of the wall rock samples is not significantly different to the mean direction of the combined kimberlite and inclusion samples, *even though* the precisions are significantly different.) Therefore the directions of all the samples have been combined to give the overall mean direction of magnetization for the De Beers pipe at the levels sampled (Table 3).

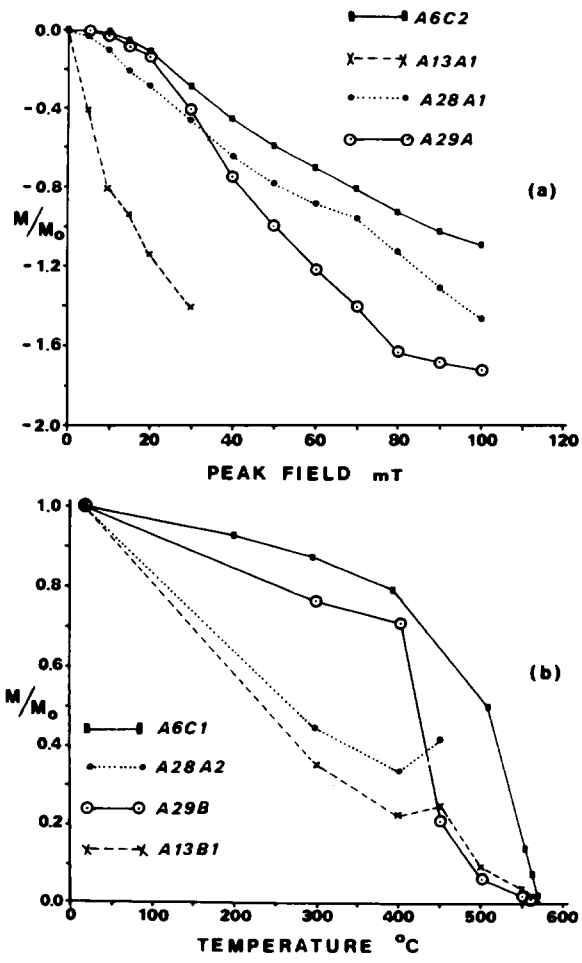


Fig. 1. Demagnetization curves for the De Beers accidental inclusion samples: (a) alternating field (vertical scale logarithmic – base 10), and (b) thermal.

3.3. DuToitspan Mine

The (almost identical) mean directions of magnetization of the kimberlite samples after application of both demagnetization techniques are given in Table 2. In both cases the value of k decreased on demagnetization which suggests the decrease is not a function of either technique but of the samples themselves. The decrease is probably due to the removal of viscous remanence acquired in the earth's present field.

The accidental inclusions collected from this mine involved such a wide range of lithologies [Karoo dolerite (4), Karoo basalt (1), fine-grained Karoo sandstone (1), highly altered olivine basalt (1), granite

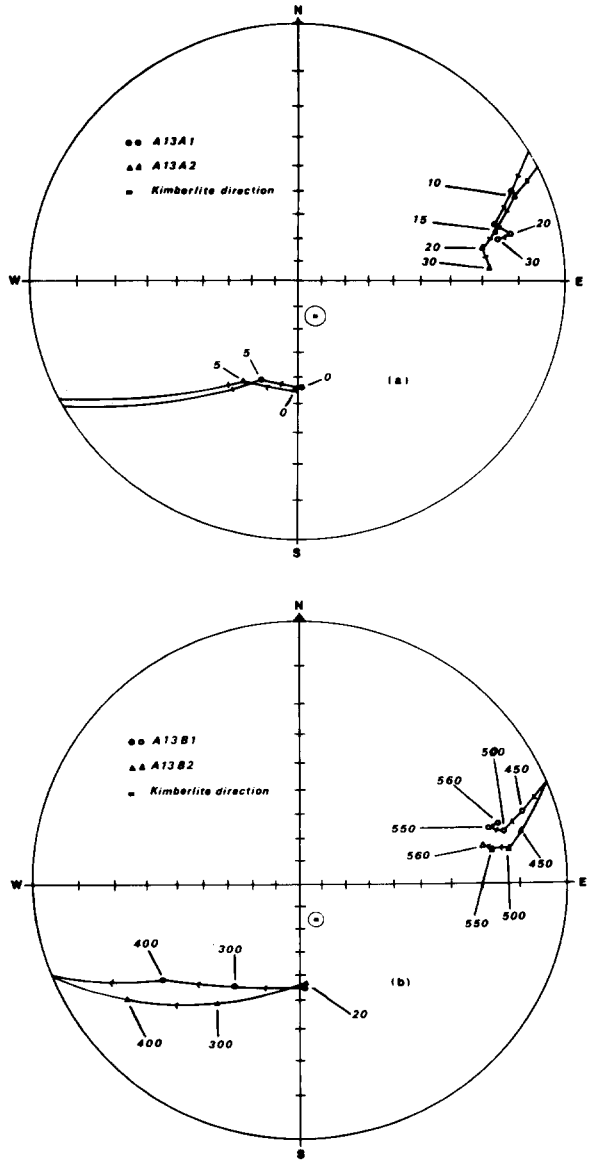


Fig. 2. Directions of magnetization during demagnetization of specimens from an accidental inclusion from De Beers Mine: (a) alternating field (values in mT) and (b) thermal (°C) Lower-hemisphere stereographic projections. Open symbols, north-seeking pole down; solid symbols, south-seeking pole down.

gneiss (1), microgranite (2)] that the use of pilot specimens was precluded. Specimens from each sample were therefore demagnetized progressively by both the thermal and magnetic methods. Four samples con-

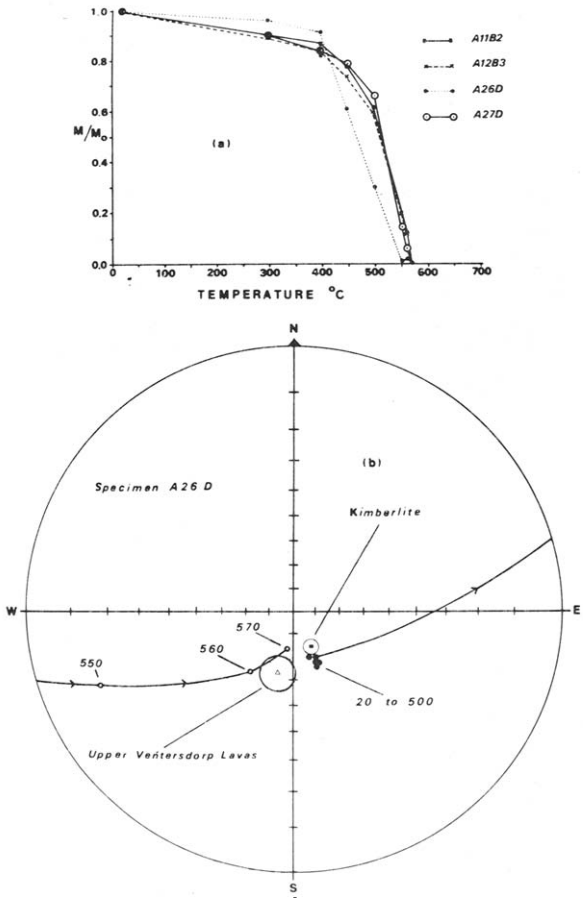


Fig. 3. (a) Thermal demagnetization curves for De Beers class 3 wall rock samples. (b) Directions of magnetization during demagnetization for one such specimen. Plotting convention as in Fig. 2.

tained high percentages of water of crystailization which caused specimens to swell and crumble at 400–450°C. For nine of the samples no changes in direction were observed during demagnetization procedures. Sample B14 behaved like A26 with the primary remanence being retained in grains with blocking temperatures above 500°C.

The wall rock samples all fell into class 2.

The mean directions of the kimberlites and accidental inclusions cannot be distinguished from each other and combination of all the sample directions gives the mean direction for the DuToitspan pipe (Table 3).

3.4. Wesselton Mine

Statistics associated with the kimberlite and accidental inclusions (all Karoo dolerite) are given in Table 2. All of the wall rock samples fall into class 2. The overall direction of magnetization for this pipe deduced from the kimberlite and inclusions is given in Table 3.

3.5. Finsch Mine

The kimberlite samples behaved unremarkably and results are given in Table 2.

The behaviour of the accidental inclusions [Karoo dolerite (6), Karoo basalt (1)] provides evidence that the magnetization of the kimberlite is primary. Progressive AF demagnetization was performed on a specimen from each inclusion (two from C1). Fig. 4a shows C14B2 exhibited large changes in direction of magnetization until a poorly defined end-point was attained. Large changes in direction were observed in the two specimens from C1; there is a large measure of consistency in their behaviour. The change in direction in C10A1 before a good end-point was achieved was much smaller in magnitude while no significant changes in direction were observed for the remaining four specimens. The direction of magnetization before and after demagnetization in 60 mT are randomly distributed.

A specimen from each inclusion was thermally demagnetized and, apart from C14B1, no large changes in direction were observed (Fig. 4b). The behaviour of C14B1 largely reproduced that of C14B2 with the direction of the removed vector between NRM and 500°C being $D = 339.5°$, $I = -62.2°$ which is close to the kimberlite mean direction. The directions before and after demagnetization in 500°C are randomly distributed. Fig. 5 shows thermal demagnetization curves for the seven specimens. The blocking temperature spectra for C1B1, C10B, C11B1 and C12A1 are almost entirely thermally discrete while those for C9A2, C13A2 and C14B1 are much more thermally distributed with the curve for C14B1 being slightly higher particularly at the upper temperatures.

There is a correlation between blocking temperature spectra and the directions of magnetization encountered during demagnetization. The blocking

TABLE 3

Mean directions of magnetization and pole positions for the five pipes

N	Treatment	D	I	k	α_{95} (deg.)	ϕ (deg.)	South poles			
							Long. (°E)	Lat. (°S)	K	A_{95} (deg.)
De Beers Mine										
18	400°C	334.1	−72.1	156.4	2.8	7.1	49.9	56.1	52.4	4.8
	(20 mT	334.6	−71.9	114.2	3.2	6.9)				
Du Toitspan Mine										
17	400°C	335.9	−69.1	81.1	4.0	4.1	53.5	59.9	35.1	6.1
	(40 mT	339.6	−70.6	71.4	4.3	5.3)				
Wesselton Mine										
8	400–463°C	337.1	−71.7	97.0	5.7	6.5	47.6	57.3	37.9	9.1
	(40–60 mT	335.5	−72.3	154.5	4.5	7.2)				
Finsch Mine										
7	400°C	315.1	−46.2	80.7	6.8	23.5	103.3	50.6	71.1	7.2
	(20 mT	312.9	−47.4	116.5	5.6	23.1)				
Koffyfontein Mine										
7	400–566°C	352.1	−75.9	39.1	9.8	11.1	30.6	54.1	13.7	16.9
	(40 mT	333.9	−78.3	45.7	9.0	12.9)				

ϕ is the angular distance between the mean direction and the direction of the earth's present magnetic field.

temperature spectra of samples C9 and C13 suggest these would be the most seriously affected by a low-temperature thermal event: their directions remained at all times close to the kimberlite direction. C12, C10 and C11 are the least likely to be affected and these have fairly stable directions each different from the kimberlite direction and from each other. C14 whose spectrum is intermediate between those of these two groups clearly shows the effect of emplacement but it was not severe enough that it could not be removed by demagnetization techniques. The behaviour of C1 on thermal demagnetization falls into the same pattern but on AF demagnetization it does not. This latter anomalous behaviour is not understood. Nevertheless the bulk of the evidence favours the correlation and it appears that the secondary component of magnetization is related to the blocking temperature spectra.

Even C9 and C13 may well contain some element of their primary directions and so the best estimate of the overall direction for the Finsch pipe has been taken as the mean of the kimberlite directions only, excluding the accidental inclusions.

3.6. Koffyfontein Mine

AF demagnetization of one specimen per kimberlite sample revealed no significant changes in direction. The demagnetization curve for E4A, however, indicates the presence of a reversed component of magnetization (Fig. 6a). Thermal demagnetization, during which no significant changes in direction occurred except for E4C1, indicated the presence of this component in each sample (Fig. 6b). Fig. 7 shows the directions of magnetization of E4C1 for various temperatures and there is clear evidence here of the presence of a roughly reversed component. This specimen was remagnetized by cooling in a known field from just above the Curie temperature of magnetite and then re-demagnetized. Data from this experiment gave no evidence to support a suggestion of partial self-reversal. At 566°C it is clear that the approximately reversed component has been removed and this temperature has been taken as the optimum demagnetizing temperature for the purpose of statistics (Table 2).

Two accidental inclusions were of Karoo dolerite

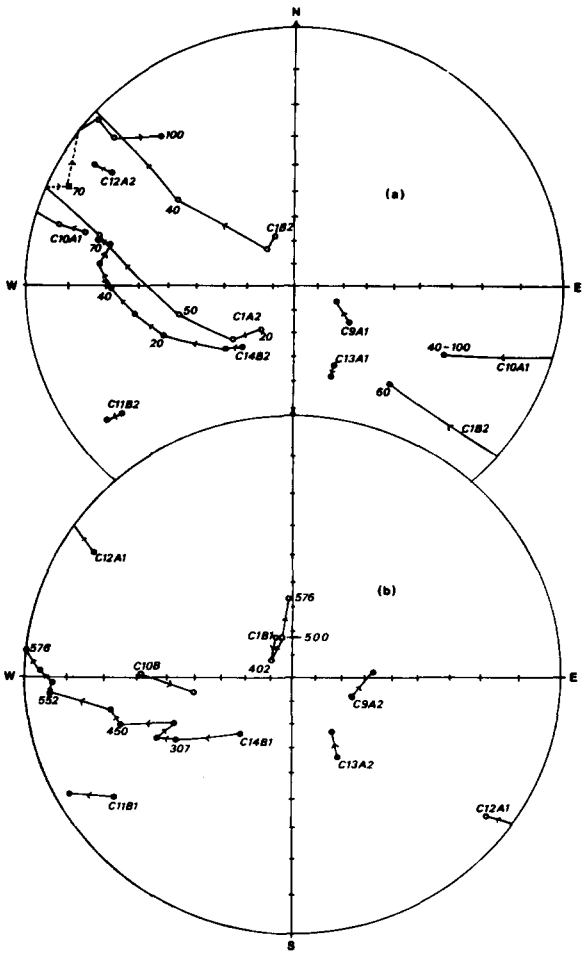


Fig. 4. Directions of magnetization during demagnetization of specimens from Finsch accidental inclusions. (a) Alternating field; directions shown are NRM and at 60 mT except where indicated otherwise. (b) Thermal; NRM and at 500°C except where indicated otherwise. Plotting convention as in Fig. 2.

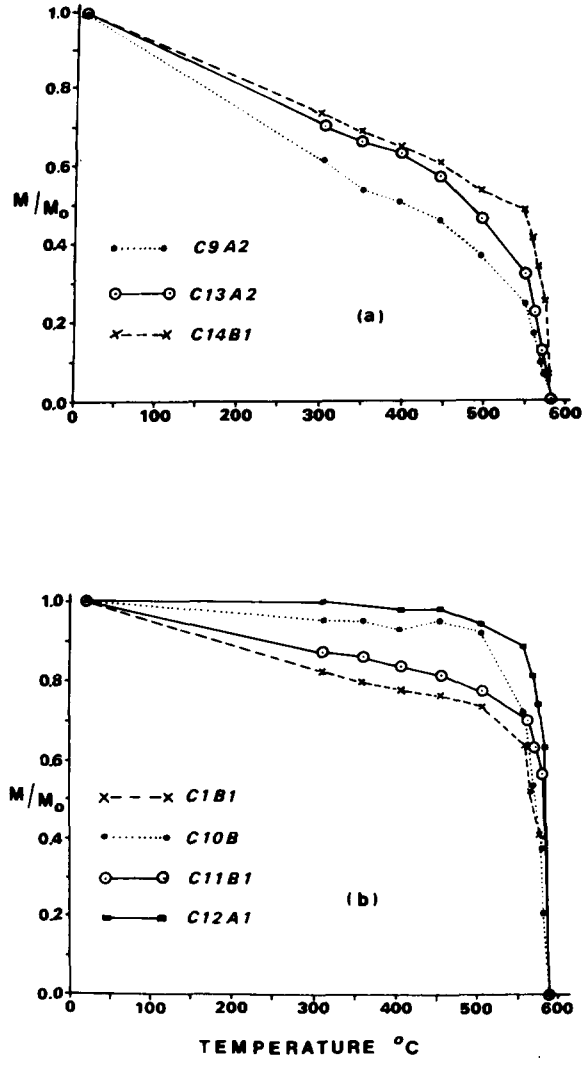


Fig. 5. Thermal demagnetization curves for the Finsch accidental inclusion samples.

while the third (E7) was of basement granite. AF demagnetization of a specimen from each revealed only an initial change in direction of about 20° in 5 mT for E7A2. Thermal demagnetization produced no changes in direction in any of the specimens and the mean direction of magnetization at 400°C is given in Table 2. The AF demagnetization curve for E7A2 and the thermal curve for E6B both indicate the presence of a reversed component of magnetization. The presence of this component in the accidental inclusions as well as the kimberlite samples argues against partial self-reversal and at present its origin is not understood. The mean directions of the kimberlite and inclusion samples after thermal demagnetization are not significantly different and all these samples have been combined to give the overall direction of magnetization for the Koffyfontein pipe (Table 3).

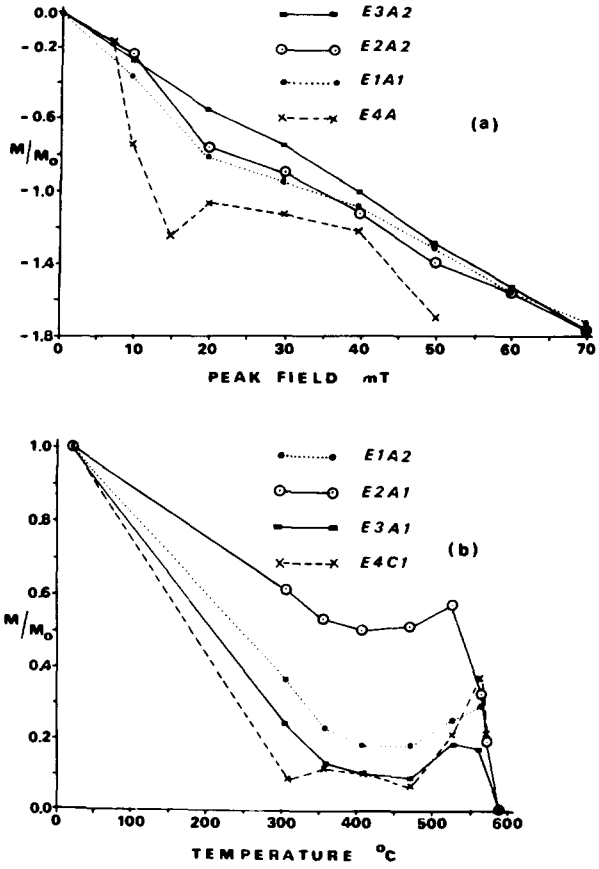


Fig. 6. Demagnetization curves for the Koffyfontein kimberlite samples: (a) alternating field (vertical scale logarithmic – base 10), and (b) thermal.

4. Primary origin of the magnetization

Kimberlite is a very basic mica-olivine rock in which the olivine has been heavily altered to serpentine. Both primary and secondary magnetite can be observed in the kimberlite [8]. Much of the secondary has resulted from the serpentinization of the olivines which is said to have occurred [8,21] contemporaneously with the emplacement of the kimberlite, or very shortly after. However, since the directions of magnetization of the pipes are similar to that of the earth's present field (e.g. at Kimberley, $D = 340°, I = -65°$) the question arises whether there has been recent serpentinization or other alteration causing a strong secondary magnetization to overprint the primary magnetization.

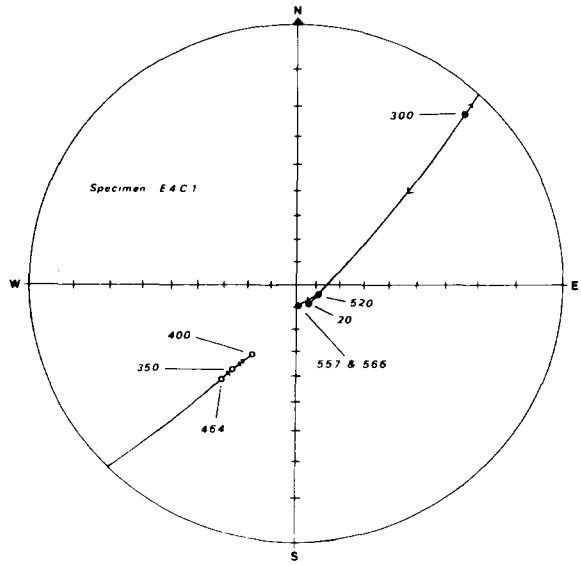


Fig. 7. Directions of magnetization during demagnetization for a kimberlite specimen from Koffyfontein Mine. Plotting convention as in Fig. 2.

We believe the observed directions are primary for the following reasons:

(1) Although the magnetization and earth's field directions appear to be similar they are in fact significantly different. Table 3 gives the angular distance between the two directions for each pipe.

(2) The mean direction of the accidental inclusions from the De Beers, DuToitspan, Wesselton and Koffyfontein Mines is, in each case, indistinguishable from the kimberlite direction. The inclusions involve a wide range of lithologies and magnetic characteristics. That inclusions of one type should have the same direction as the matrix is evidence for the originality of the magnetization; such a wide variety of types is even more convincing. In addition petrographical examination of all the inclusions shows no evidence of any recent alteration.

(3) The mean directions of class 3 wall rock samples on or very close to the contact, are indistinguishable from the appropriate kimberlite directions. Further from the contact samples from the same rock type give the appropriate primary direction with little evidence of a secondary overprint. There is clearly a thin baked contact zone.

(4) The secondary component of magnetization in the Finsch accidentals is related to the blocking tem-

perature spectra implying a thermal rather than a chemical origin.

(5) Fresh kimberlitic material collected from deep underground in the Premier Mine [7] has been shown to be capable of retaining a Precambrian direction of magnetization. Further R.B. Hargraves (personal communication), who has studied other southern African kimberlites, has found that one of the three late Jurassic fissures near Swartruggens is reversely magnetized while the National pipe, near the Premier Mine, has retained its Precambrian direction of magnetization.

5. Discussion

The mean (south) pole position for each pipe is given in Table 3. The poles are obviously not virtual geomagnetic poles (representing an instant in time). In the following, however, it is suggested that none of them should be regarded as full palaeomagnetic poles in which secular variation has been completely averaged out.

Kimberlitic material emplaced at 350°C (the upper limit deduced by McFadden [11] for the levels sampled) into a pipe of similar dimensions to that of De Beers would cool to 40°C in only 430 years [10,22]. Thus if TRM was the only mechanism involved secular variation would not have been averaged out. However, much of the magnetization of the kimberlite itself is likely to have a CRM origin (early alteration of the olivines) and here it is difficult to estimate the time scale involved. The circular standard deviation (angular dispersion) of directions $\Delta = 81/k^{1/2}$ is the most usual measure of the palaeosecular variation. For De Beers, $\Delta = 6.5°$ with 95% confidence limits of 5.2° and 8.5° (calculated using the procedure outlined by Cox [23]). This value is thus significantly less than the expected value of 10.6° for a palaeolatitude of 57.1° [24]. This implies that cooling in the De Beers pipe *and* significant serpentinization of the olivines had terminated in a time less than that required to average out secular variation.

The relevant figures for DuToitspan are $\Delta = 9.0°$ (7.2° and 11.9°) against an expectancy of 11.0° and for Wesselton, $\Delta = 8.2°$ (6.0° and 13.0°) against 10.7°. However, because both pipes are larger than De Beers (thus lower values of k are to be expected) and because the number of samples in each case is not very large (this reduces confidence in k) the conclusion that secular variation has been averaged out in these pipes is not necessarily correct. It is evident from Table 3 that the mean direction of the three Kimberley pipes are not just "not significantly different" (which can be confirmed statistically) but are almost identical (k for the three directions is 1056) This and the fact that these are small, rare geological formations lying within two or three kilometers of each other, suggests quite reasonably that they were emplaced almost simultaneously. Combining all the samples from these mines gives $k = 107.1$ with $\Delta = 7.8°$ (6.8° and 9.2°) and an expectancy of 10.8°. This implies that the De Beers, DuToitspan and Wesselton pipes were all emplaced, cooled and significant serpentinization of the olivines had terminated in a time less than that required for secular variation to average out. Hence the same must be true for the DuToitspan and Wesselton pipes individually.

For Koffyfontein $\Delta = 13.0°$ (9.3° and 21.4°); here the scatter cannot be attributed solely to secular variation but involves also the movement of samples during mining operations. Clearly it is fallacious to compare these figures with an expectancy of 10.1° and conclude that secular variation has been averaged out.

The Finsch pipe has the largest cross-section and the time for emplaced material to cool from 350°C to 40°C would be close to 3900 years. The upper confidence limit of 14.9° on Δ (9.0°) just exceeds the expected value of 14.7°. This implies a slightly greater than 5% chance that secular variation has been averaged out. Such marginal significance suggests it might be wiser not to consider Finsch as a full palaeomagnetic pole.

Whichever way an overall pole position is calculated (and it is not easy to decide which way is correct) the result is not very different. The mean (south) pole for the five pipes is at 58.2°S, 57.2°E with $K = 25.8$ and $A_{95} = 15.3°$. If, as a result of the inferred simultaneous emplacement of the Kimberly pipes, it is concluded that only three independent measurements have been made then the mean pole is at 58.3°S, 62.2°E with $K = 13.6$ and $A_{95} = 34.6°$. The angle between the two means is only 2.6°. The 57 individual sample poles give 58.1°S, 55.1°E with $K = 21.7$ and $A_{95} = 4.1°$. For each of these mean poles a value of Δ can be calculated and

compared with that to be expected [25] as a result of secular variation (16°). On any of the combinations the data give no reason to assert that secular variation has not been averaged out and it is believed that any of the above provide a good estimate of the palaeomagnetic pole position 86 ± 3 Ma ago.

Recently McElhinny and Brock [26] have reviewed palaeomagnetic results from Mesozoic rocks in Africa including that from the kimberlite pipes. It is sufficient to state here that the African palaeomagnetic pole remained essentially in the same position from early Triassic (Maji ya Chumvi) to late Cretaceous (kimberlites, Nubian sandstones, etc.).

The Monastery pipe of the eastern Orange Free State has also been dated at 86 ± 3 Ma [6] and studied palaeomagnetically by R.B. Hargraves (personal communication, manuscript in preparation) together with five pipes in nearby Lesotho (Kao, Letseng, Lipalaneng, Marakabei and Pipe 200). He reports that their directions of magnetization are all very similar with a mean direction very close to that of the five pipes of the present study. Clearly a widespread kimberlite event occurred in southern Africa some 86 ± 3 Ma ago and it is not unreasonable to relate the origin of forces causing crustal failure deep enough to allow widespread injection of kimberlite pipes to the onset of plate motions involving Africa. It is difficult to envisage any other mechanism which could cause forces of the required magnitude dispersed over a substantial portion of southern Africa.

Acknowledgments

This study was undertaken during the tenure by one of us (P.L.McF.) of a Beit Fellowship and a Three Feathers Scholarship and this support is gratefully acknowledged. We thank the staff of De Beers Consolidated Mines Limtited for their assistance during sampling, A.B. Reid for helpful discussions throughout the study and A. Brock and R.B. Hargraves for critical reading of the manuscript. The latter kindly provided us with data in advance of publication.

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